Maternal Vitamin D and Gene Expression in Normal Pregnancy and Preeclampsia

HSE 103 Final Project | Fall 2026

Background and research question

Preeclampsia is a pregnancy complication that can affect both the mother and the baby. I was interested in whether vitamin D levels might be related to changes in gene expression during early pregnancy. The original study looked at vitamin D, preeclampsia, and blood gene expression in a much larger project called VDAART (Mirzakhani et al., 2016). I used the subset of data provided for this course to look at a few genes in more detail.

My main question was: How does expression of *CXCR2*, *CYP24A1*, and *PTX3* vary with maternal vitamin D concentration and pregnancy outcome?

Methods

Data and participants

I used the three original course files: GSE85307_covar-1.csv, GSE85307_geneExpression.csv, and GSE85307_geneNames.csv. The data are linked to GEO series GSE85307 and the study by Mirzakhani and colleagues (2016). The original VDAART trial enrolled 881 women, but this particular gene-expression analysis is a nested sample of 157 women. In the files I analyzed, 110 had normal pregnancies and 47 had preeclampsia. Blood samples were taken around 10–18 weeks of pregnancy, and gene expression was measured using an Affymetrix Human Gene 1.0 ST microarray (NCBI GEO, GSE85307).

All 157 participants in the provided dataset were included. I did not exclude any rows when preparing the variables used in this report. I used each sample's geoID to match the clinical information to the gene-expression matrix, then used the gene-name linker file to identify the appropriate probe for each gene.

Variables and analysis

Maternal vitamin D was used both as a continuous value in ng/mL and as two groups: below 30 and 30 or higher. Pregnancy outcome was coded as Normal or Preeclampsia. I also summarized maternal age and BMI. The expression values are displayed as log2 signal, as in the supplied dataset and class plots.

For each of *CXCR2*, *CYP24A1*, and *PTX3*, I made a histogram, a scatterplot with a LOESS smooth curve, and a grouped boxplot. Table 1 compares the two pregnancy-outcome groups using means and standard deviations or counts and percentages. I also made a heatmap of 10 available genes, with hierarchical clustering of both genes and participants and two colored tracking bars for pregnancy outcome and vitamin D group. The heatmap shows unstandardized expression values. Finally, the lollipop plot shows the mean expression difference for each of those 10 genes (preeclampsia minus normal).

These are exploratory, descriptive comparisons. I did not run significance tests or adjust for other variables, so the plots alone cannot establish a causal or statistically significant relationship.

Software and reproducibility

I used R version 4.5.1 with RStudio and R Markdown to prepare the data and create the figures (R Core Team, 2025). For data preparation, I used tidyverse 2.0.0 (Wickham et al., 2019), including dplyr 1.2.1 (Wickham et al., 2026a) and tidyr 1.3.2 (Wickham et al., 2026b). I used ggplot2 3.5.2 for the plots (Wickham, 2016), pheatmap 1.0.13 for the heatmap (Kolde, n.d.), and knitr 1.50 to generate the R Markdown report (Xie, 2014).

GitHub repository: https://github.com/rayrayxie/HSE103_FinalProject

Results

Participant characteristics

Table 1 shows that the average maternal age was similar between the groups: 26.63 years for normal pregnancies and 25.78 years for preeclampsia. BMI was higher in the preeclampsia group (31.16 vs. 27.99 kg/m^2). Mean vitamin D was lower in the preeclampsia group (19.72 vs. 24.37 ng/mL). In addition, 87.2% of women with preeclampsia were below 30 ng/mL, compared with 70.9% of women in the normal group. These are differences in this sample, not adjusted estimates of risk (Table 1).

Variable	**Normal (n = 110)**	**Preeclampsia (n = 47)**
Maternal age (years)	26.63 (5.08)	25.78 (4.95)
Maternal BMI (kg/m^2)	27.99 (7.66)	31.16 (8.03)
Vitamin D (ng/mL)	24.37 (14.55)	19.72 (8.33)
Vitamin D below 30 ng/mL	78 (70.9%)	41 (87.2%)
Vitamin D 30 ng/mL or higher	32 (29.1%)	6 (12.8%)

Table 1. Maternal characteristics by pregnancy outcome. Values are mean (standard deviation) for continuous variables and n (%) for categorical variables.

Gene expression figures

The next nine plots show the distribution of each selected gene, how its expression varies across vitamin D values, and how it looks within the pregnancy-outcome and vitamin D groups. I kept the same plot types for all three genes so the comparisons are easier to follow.

CXCR2

The histogram for *CXCR2* shows most expression values near the upper part of its observed range, with a few lower values (Figure 1). The scatterplot has a slightly curved trend, but there is a lot of overlap in the points, particularly at lower vitamin D values (Figure 2). In the grouped boxplot, the vitamin D 30-or-higher groups tend to have higher medians, although the boxes overlap (Figure 3).

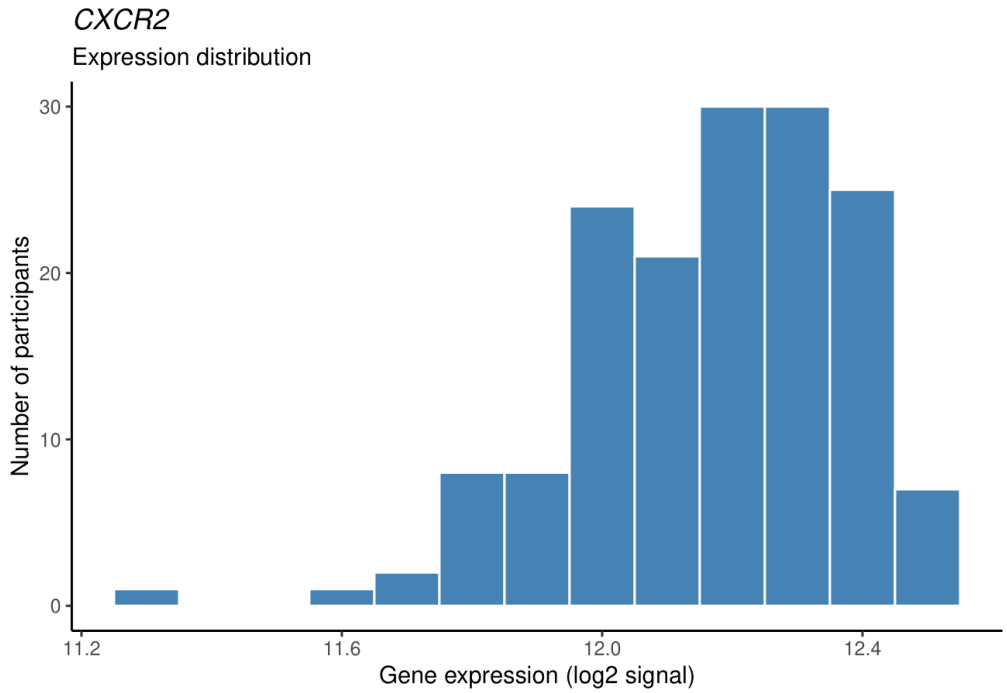


Figure 1. Distribution of *CXCR2* expression across the 157 participants.

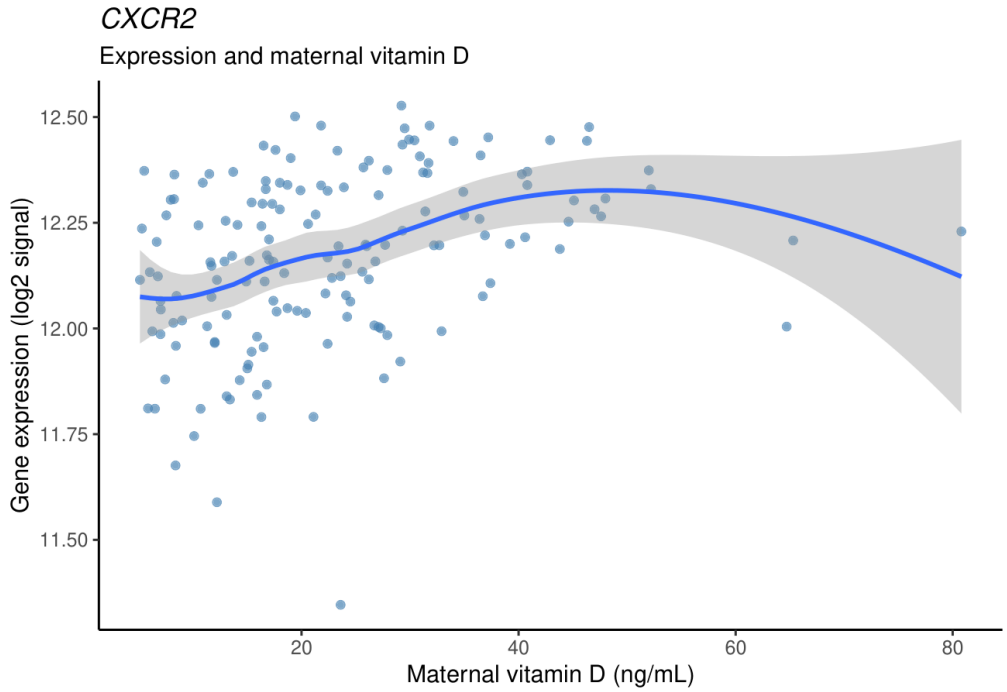


Figure 2. *CXCR2* expression and maternal vitamin D. The blue curve is a LOESS smooth; the shaded area reflects uncertainty around the fitted curve.

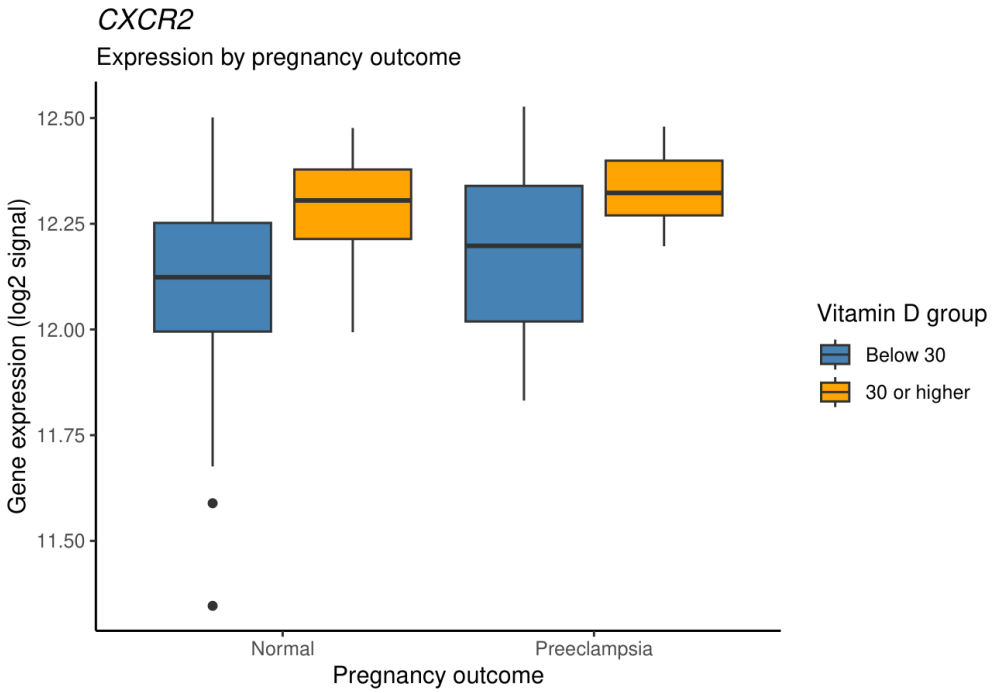


Figure 3. *CXCR2* expression by pregnancy outcome and vitamin D category. The boxplots show the median, middle 50% of values, and outliers.

CYP24A1

The *CYP24A1* histogram is fairly concentrated, with most values around 4.7–4.9 (Figure 4). Its scatterplot does not show a clear steady increase or decrease as vitamin D rises (Figure 5). The boxplot medians are close together across the different groups, and the distributions overlap quite a bit (Figure 6).

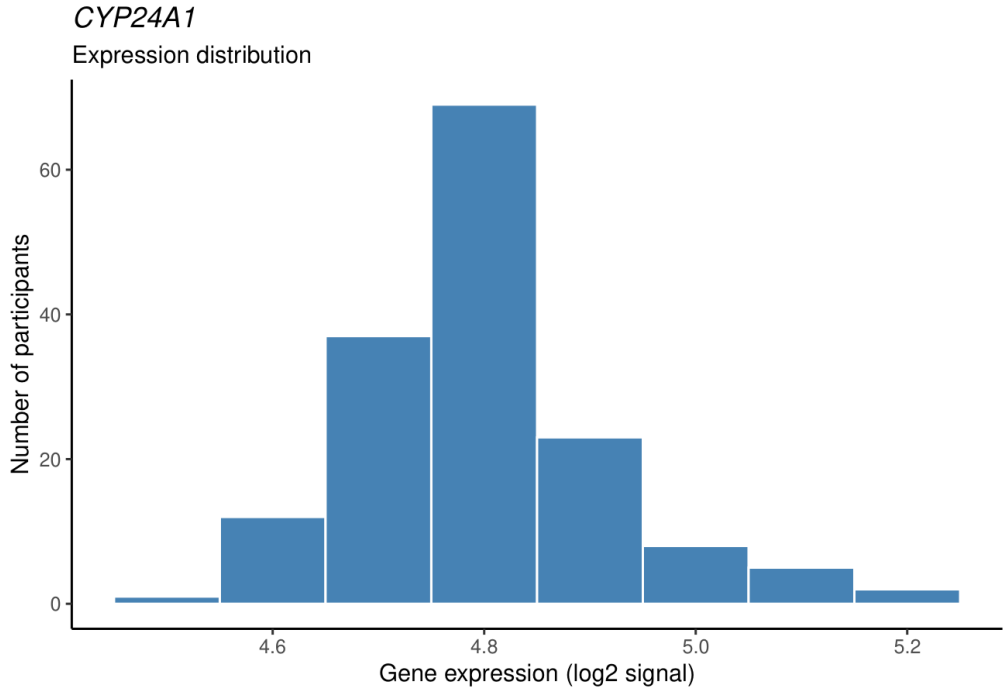


Figure 4. Distribution of *CYP24A1* expression in the study sample.

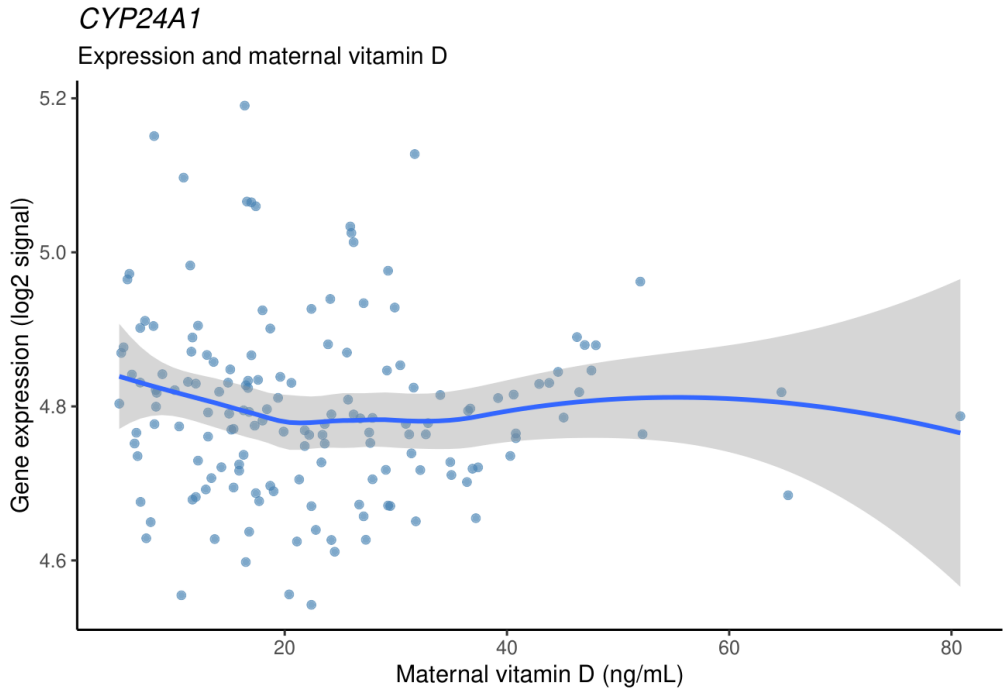


Figure 5. *CYP24A1* expression versus maternal vitamin D with a LOESS smooth curve.

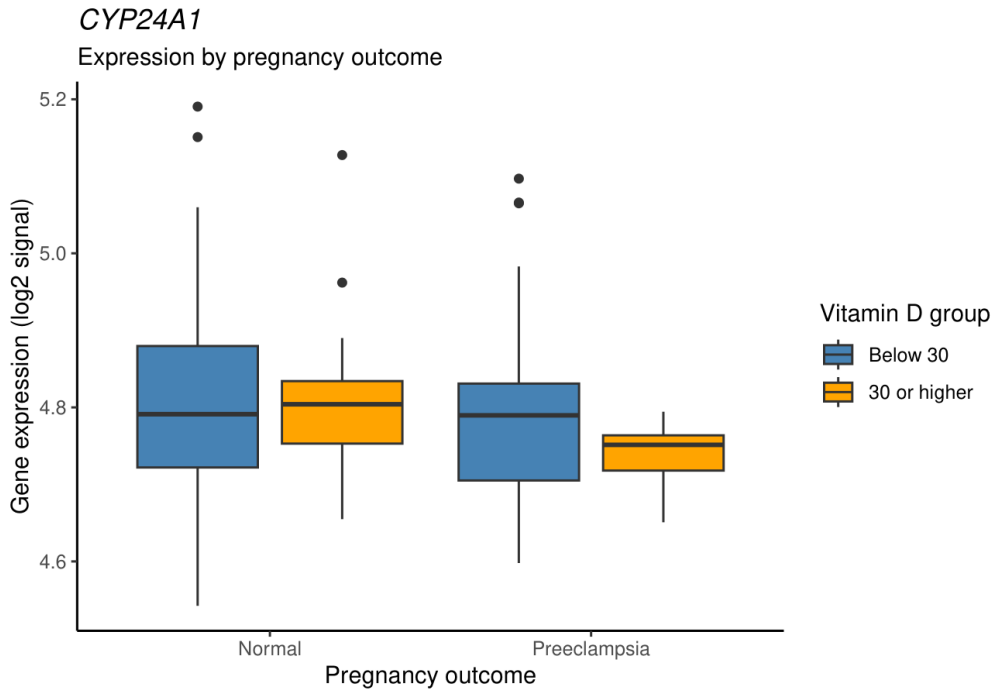


Figure 6. *CYP24A1* expression separated by pregnancy outcome and vitamin D group.

PTX3

The *PTX3* expression values are mostly grouped near 4.8–5.0, although a few participants have noticeably higher values (Figure 7). The scatterplot looks mostly flat to me, with a little curvature, but not a strong pattern (Figure 8). Its grouped boxplots overlap between normal pregnancies and preeclampsia as well (Figure 9).

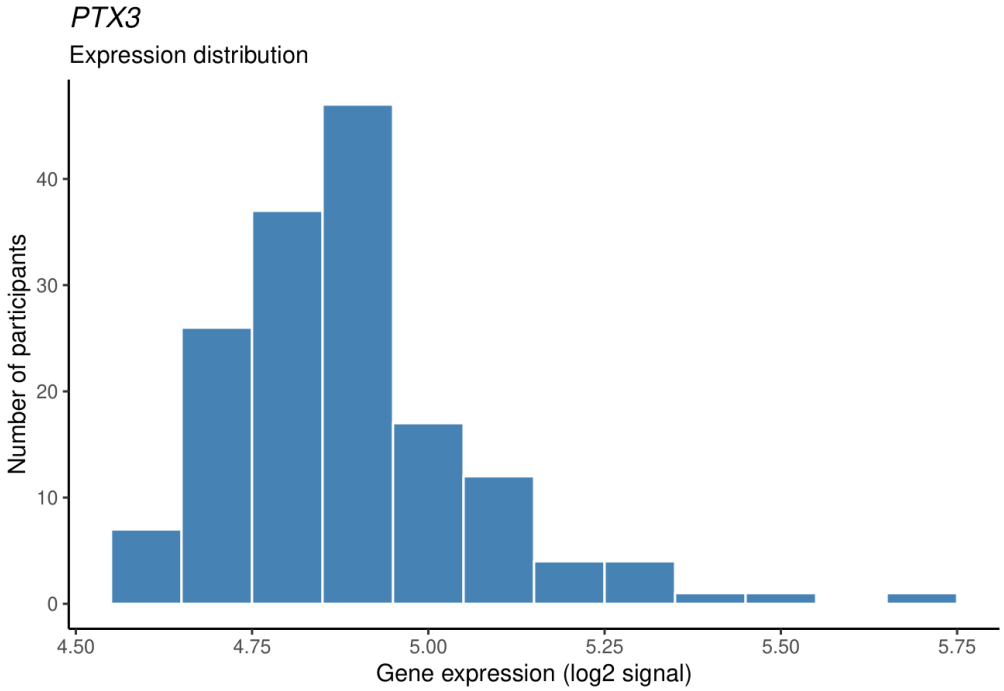


Figure 7. Distribution of *PTX3* expression; a small number of higher values are visible.

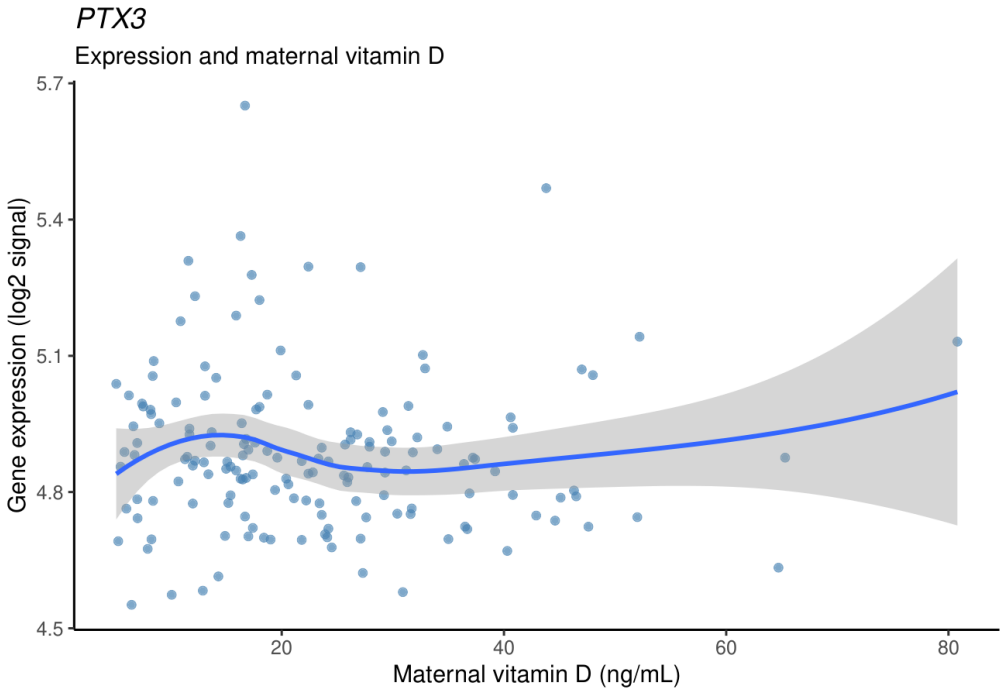


Figure 8. *PTX3* expression and maternal vitamin D with a LOESS smooth curve.

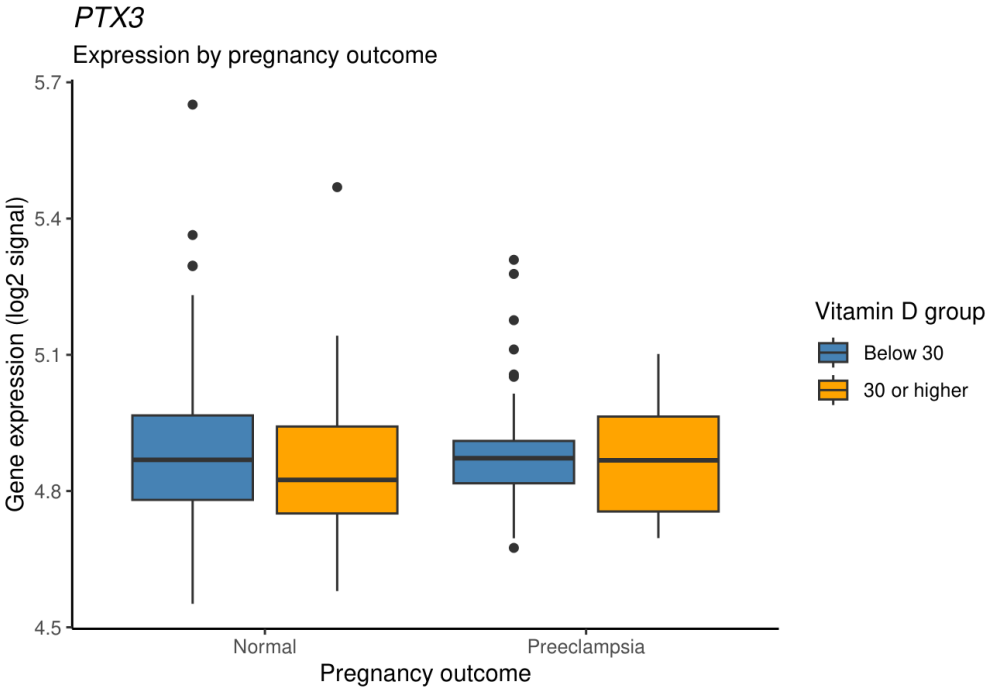


Figure 9. *PTX3* expression by pregnancy outcome and vitamin D category.

Ten-gene heatmap

For the heatmap I included *CXCR2*, *CYP24A1*, *PTX3*, *SOCS5*, *SIRT3*, *HSPA8*, *CNR1*, *TRIM24*, *SLC1A5*, and *RBL2*. The clustering groups genes and participants according to their expression patterns (Figure 10). The gene rows show different overall baseline expression levels, which explains a lot of the red-versus-blue coloring. The colored pregnancy-outcome and vitamin D bars are mixed across the columns rather than forming one obvious block for either group. Since the values were not standardized across genes, I would be careful about interpreting this plot as evidence of a biological cluster.

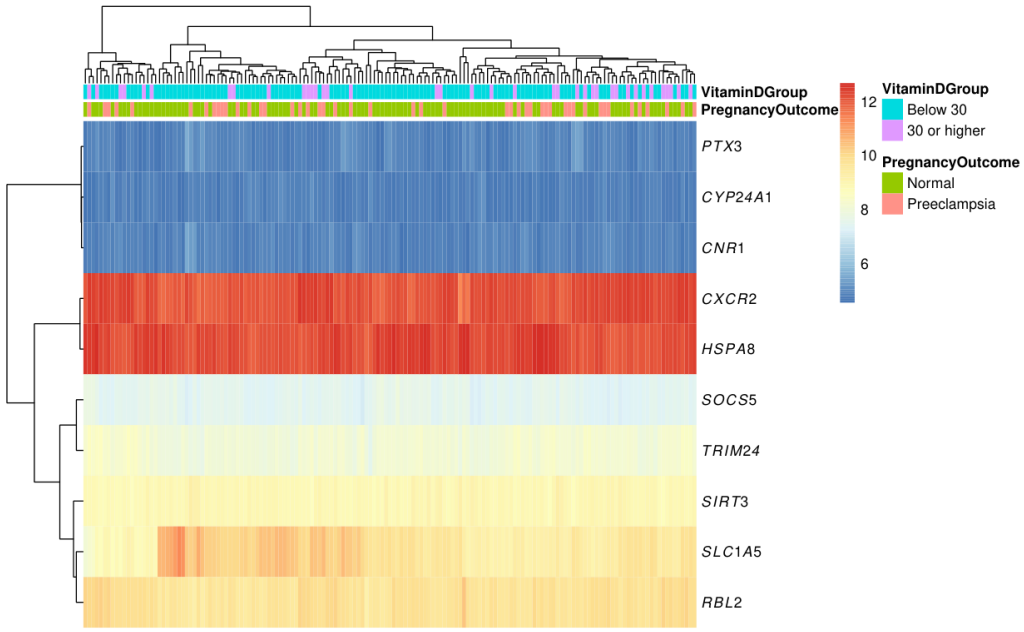


Figure 10. Heatmap of 10 genes across 157 samples, clustered using Euclidean distance. Annotation bars show pregnancy outcome and vitamin D category.

Additional visualization: lollipop plot

The lollipop plot compares the mean expression in preeclampsia minus the mean in normal pregnancies (Figure 11). Values to the right of zero are higher on average in preeclampsia, and values to the left are lower. In this sample, SLC1A5 has the largest negative difference (about -0.047), while *CXCR2* and TRIM24 show the largest positive differences (about +0.037 each). Most values are fairly close to zero. This is a useful quick comparison, but it does not tell us whether the differences are statistically significant.

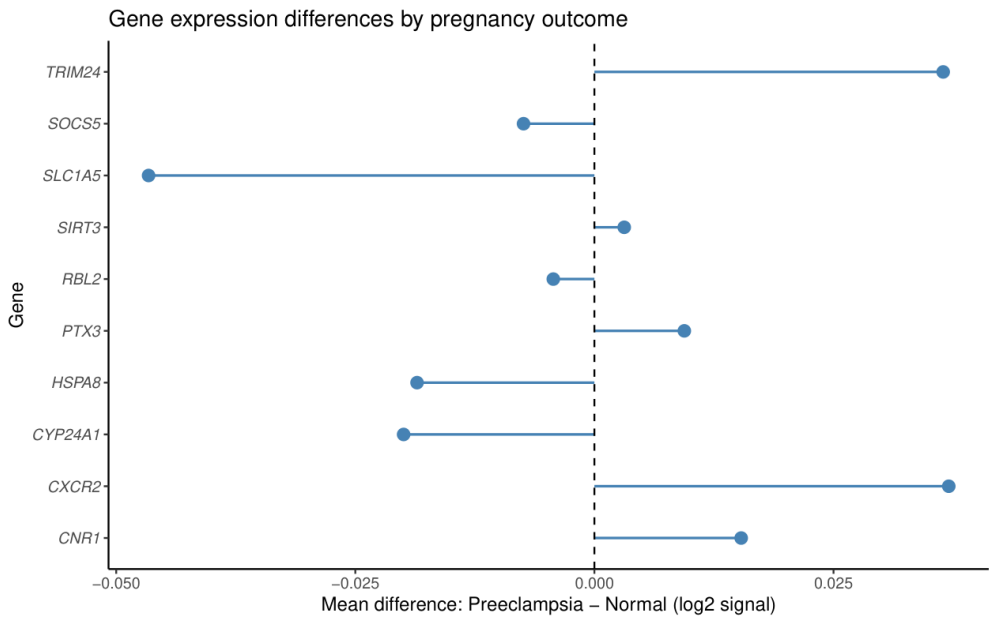


Figure 11: Difference in average log2 gene-expression signal (preeclampsia minus normal) for the 10 selected genes.

Short discussion

Overall, the clearest clinical difference in Table 1 was that women in the preeclampsia group had lower average vitamin D and higher BMI. The gene plots are less clear-cut. I can see some small trends, but there is a lot of overlap, and the heatmap does not separate the outcome groups into obvious clusters. I think the lollipop plot is useful because it puts the 10 genes on the same comparison scale without having to look through 10 separate boxplots.

A main limitation is that this is a small selected sample from a larger study, and the group sizes are uneven. Only six participants in the preeclampsia group had vitamin D at or above 30 ng/mL. I also did not adjust for maternal age, BMI, batch, or other possible confounders. The heatmap uses raw log2 expression rather than row-standardized values, so baseline differences between genes can dominate the colors. I would treat this as an initial look at the data rather than a final biological conclusion.

References

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